Data Flow Diagram

Date	27 June 2026
Team ID	-
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	Represents the user interacting with the VBCUA system.
Rectangle with numbered header	**Process**	Performs operations such as speech transcription, semantic analysis, keyword matching, and score generation.
Rectangle (solid fill, no number)	**Data Store**	Stores reference concepts, analysis history, reports, and database records.
Labeled arrow	**Data Flow**	Represents the movement of audio, transcripts, scores, and reports between processes

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.


Student
Upload Audio File
1. Audio Upload & Validation
Valid Audio File
2. Speech-to-Text
(Faster Whisper)
Transcript
3. Semantic Similarity
(Sentence-BERT)
4. Keyword Matching
5. Score & Feedback Generation
Overall Score / Feedback
6. Save to SQLite Database
PDF
7. Generate PDF Report
Dashboard & User Results